tube or, even more realistically, like a thin three-dimensional doughnut? The seemingly insurmountable difficulties found by Heisenberg, Dirac, and others in their attempts to construct a quantum theory of three-dimensional nuggets have repeatedly stymied researchers following this natural chain of reasoning.

Quite unexpectedly, though, during the mid-1990s, string theorists realized, through indirect and rather shrewd reasoning, that such higher-dimensional fundamental objects actually do play an important and subtle role in string theory itself. Researchers have gradually realized that string theory is not a theory that contains only strings. A crucial observation, central to the second superstring revolution initiated by Witten and others in 1995, is that string theory actually includes ingredients with a variety of different dimensions: two-dimensional Frisbee-like constituents, three-dimensional blob-like constituents, and even more exotic possibilities to boot. These most recent realizations will be taken up in Chapters 12 and 13. For now we continue to follow the path of history and further explore the striking new properties of a universe built out of one-dimensional strings instead of zero-dimensional point-particles.